The Charged MMST Scaling-Limit Theorem: A Complete In-House Proof Attempt for `SEAM.MMST.TYPEIII.CHARGED.01`

Proof memorandum

30 August 2026

Abstract

We attempt the complete scaling-limit theorem for the concrete order-four charged field

$$\Psi_{\lambda,N}(f) = \varepsilon_\lambda :\exp(i\lambda \cdot \phi_N): (f), \qquad \lambda = (\omega_s, \omega_f), \quad \|\lambda\|^2 = 2, \quad h_\lambda = 1.$$

The algebraic limit-exchange statement is proved in full for an arbitrary finite group: equivariant injective inductive limits commute isometrically with crossed products, coefficient and fixed-point expectations commute with the limit, and their Pimsner–Popa constants remain uniform. Assuming the charged field and local algebras exist, the identification of the extension with the holomorphic $(E_8)_1$ net is reduced, with every hypothesis displayed, to the Kawahigashi–Longo–Müger index theorem, the rank-eight even-unimodular lattice classification, the Dong–Mason central-charge-eight theorem, and the Carpi–Kawahigashi–Longo–Weiner VOA–net correspondence.

The uniform Green-form problem is closed here for the edge-projected frozen collar, with the explicit all-volume constant $K_G = \pi^2/4$. The proof combines an exact particle–hole count, the sharp fundamental-domain sine inequality, and analytic perturbation of the isolated edge doublet. For the edge-projected boson modes we also construct exact compatible dyadic Fock isometries and prove the charged-field graph defect with rate N^{-2}, including adjoints. The remaining hard lattice-to-field step is microscopic local-algebra exhaustion. The frozen computations verify the proposed two-edge remainder bound $\|R_N\|_{\mathrm{HS}} \leq 2.11$ only for $N = 16, 24, 32, 48, 64, 96$ (maximum 2.105079358), not for all N. Consequently the master theorem remains conditional on one analytic gap box, stated quantitatively at the end. This is a full proof attempt, not a claim that finite observations establish a type-III scaling limit.

Contents

1 Verdict and logical form

Theorem 1.1 (Charged MMST master theorem: exact proved form)**.** *Let the finite collars, scaling maps, neutral current limit and charged operators be as in Section 2. Suppose the scaling algebra is exhausted by the neutral currents and the four charged grades as specified in* (ALG-EXH) *there. Then:*

(L1) *For every $f \in C^\infty(S^1)$ there are $p < \infty$ and $C_f < \infty$, independent of N, such that on the finite-energy core*

$$\|\Psi_{\lambda,N}(f)\xi\| \leq C_f \, \|(1 + L_{0,N})^p \xi\| \,. \tag{1.1}$$

(L2) *After the canonical scaling identifications, $\Psi_{\lambda,N}(f)$ converges strongly on the common finite-energy core to a nonzero closable field $\Psi_\lambda(f)$. The generated local algebras converge to a local, conformal extension net $I \mapsto \mathcal{B}(I)$ of the neutral net $I \mapsto \mathcal{A}(I)$.*

(L3) *The finite $\mathbb{Z}_4$ crossed products, the canonical coefficient expectations, and the fixed-point expectations commute with the inductive/scaling limit. The limiting Pimsner–Popa bound is still $E(x) \geq \frac{1}{4}x$ for $x \geq 0$.*

(L4) *The resulting extension is the strongly local lattice net $\mathcal{A}_{V_{E_8}}$, equivalently the level-one E_8 current net. It is holomorphic, has $\mu = 1$, central charge 8, vacuum character E_4/η^8, and local algebras equal to the hyperfinite type-*III_1 *factor.*

Proof. L1 follows from Theorem 4.8. L2 follows from Theorem 5.18. L3 is the unconditional Theorem 6.1. L4 follows from Theorem 7.1. The Green estimate used by L1 is the all-volume Theorem 4.7; the graph estimate used by L2 is the all-volume dyadic Theorem 5.3. The condition still used by L2 is precisely (ALG-EXH) in the final gap box; it is not silently replaced by a finite-volume fit. □

Remark 1.2 (What is and is not proved). The implication in Theorem 1.1 is proved. Uniform Green domination and local Sobolev convergence are proved in Theorem 4.7, and edge-mode graph convergence is proved in Theorem 5.3. The antecedent (ALG-EXH) is not proved for the frozen microscopic collar. Therefore this memorandum does not close `SEAM.MMST.TYPEIII.CHARGED.01`. It proves L3, completes the classical reduction of L4, closes L1 and the edge-projected TEL-a part of L2, and isolates the remaining microscopic local-algebra statement.

2 Setup: the frozen collar and charged generator

2.1 The finite CAR collar

For circumference N and fixed open width $N_y = 8$, the one-particle Qi–Wu–Zhang collar is

$$h(k_x, k_y) = \sin k_x\,\sigma_x + \sin k_y\,\sigma_y + (1 - \cos k_x - \cos k_y)\sigma_z. \tag{2.1}$$

The rth deck sector, $r \in \mathbb{Z}_4$, has boundary phase i^r, hence longitudinal momenta

$$p_n^{(r)} = \frac{2\pi}{N}\left(n + \frac{r}{4}\right). \tag{2.2}$$

Sixteen Majorana copies produce the carrier central charge $16 \cdot \frac{1}{2} = 8$. We write $\mathfrak{h}_N$ for the one-particle space, $H_N^{(r)}$ for the corresponding one-particle matrix, and

$$C_N^{(r)} = \chi_{(-\infty,0)}(H_N^{(r)})\frac{1}{2}\chi_{\{0\}}(H_N^{(r)}) \tag{2.3}$$

for the particle–hole symmetric covariance. The finite observable algebra is a matrix CAR algebra. Its low-energy GNS space is denoted $\mathcal{H}_N$, and $L_{0,N}$ denotes the shifted, nonnegative edge-energy operator in the scaling normalization.

Remark 2.1 (Necessary correction: there is no uniform full gap). The topological collar has chiral edge levels with spacing $O(N^{-1})$. Thus a claim of a spectral gap $\inf_N \operatorname{gap}(H_N) > 0$ is false. The usable hypothesis is a *uniform bulk gap after removing the edge band.* If Q_N is the edge-band projection, the correct statement is

$$\operatorname{spec}\big((1 - Q_N)H_N(1 - Q_N)\big) \cap (-\delta_{\text{bulk}}, \delta_{\text{bulk}}) = \varnothing \tag{2.4}$$

for one $\delta_{\text{bulk}} > 0$ independent of N. This controls bulk resolvents but does not control the massless longitudinal Green function by itself.

2.2 The exact order-four vector

In rational coordinates,

$$\omega_s = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right), \qquad \omega_f = \left(\frac{3}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}\right), \qquad \lambda = (\omega_s, \omega_f). \tag{2.5}$$

The exact finite calculation gives

$$\|\omega_s\|^2 = \frac{5}{4}, \quad \|\omega_f\|^2 = \frac{3}{4}, \quad \|\lambda\|^2 = 2, \quad h_\lambda = 1, \tag{2.6}$$

and

$$[\lambda]^4 = [0], \qquad [\lambda]^2 = [v]. \tag{2.7}$$

The norm-two vectors in $D_5 \oplus A_3 + k\lambda$, $k = 0, 1, 2, 3$, have census

$$[52, 64, 60, 64], \tag{2.8}$$

so the four cosets contain 240 roots and the two odd grades contain the full 128-dimensional spinor contribution.

2.3 The finite charged field

Let $\phi_N = (\phi_N^1, \ldots, \phi_N^8)$ be the bosonic current obtained from the sixteen Majorana copies by edge-projected collar bosonization, with zero mode fixed by the $\mathbb{Z}_4$ charge sector. Each complex-fermion pair gives one boson. The longitudinal derivative uses the frozen lattice symbol

$$\omega_N(m) = \frac{N}{\pi} \left| \sin \frac{\pi m}{N} \right|, \qquad 0 < |m| \leq \frac{N}{2}. \tag{2.9}$$

The edge projection is the analytic top (or bottom) eigenbranch through the zero-energy crossing; massive bulk-complement modes are not included in ϕ_N. For a fixed normalized two-cocycle ε_λ, define on the finite-energy vectors

$$\Psi_{\lambda,N}(f) = \sum_m \widehat{f}_m \, \varepsilon_\lambda \, {:}\exp(i\lambda \cdot \phi_N){:}_m \, . \tag{2.10}$$

Normal ordering is with respect to $C_N^{(0)}$. Formula (2.10) is meaningful at every finite N. The theorem does not follow merely from that fact: the normal-ordering constant and the graph norm of the exponential can grow with N.

Definition 2.2 (Scaling convergence used below)**.** *Let $V_N : \mathcal{H}_N \to \mathcal{H}_\infty$ be the canonical isometries of the chosen operator-algebraic renormalization system. A sequence of closable operators X_N converges on the finite-energy core*

$$\mathcal{D}_{\mathrm{fin}} = \bigcup_{E<\infty} \mathbf{1}_{[0,E]}(L_0)\mathcal{H}_\infty \tag{2.11}$$

if for every $\xi \in \mathcal{D}_{\mathrm{fin}}$, represented at sufficiently large scale,

$$V_N X_N V_N^* \xi \longrightarrow X\xi, \qquad V_N X_N^* V_N^* \xi \longrightarrow X^* \xi. \tag{2.12}$$

The adjoint clause is included because bare pointwise convergence of unbounded fields is too weak to determine local von Neumann algebras.

3 Classical inputs, quoted in the forms used

Theorem 3.1 (Osborne–Stottmeister free-fermion scaling theorem, form used here)**.** *For the massless chiral lattice fermion systems constructed by operator-algebraic renormalization, with the regular scaling functions and compatible ground states specified in their construction, the renormalized Koo–Saleur approximants converge strongly on the finite-particle core to the corresponding Virasoro generators. The smeared approximants converge on that core, and the same argument applies to level-one Wess–Zumino–Witten currents which are fermion bilinears. Their dynamical correlation functions converge uniformly on compact time intervals under the stated regularity assumptions on the smearing functions.*

This is the content used from Osborne–Stottmeister, *Conformal field theory from lattice fermions*, Theorems 4.11, 4.16, 5.1, 6.1 and 6.3 [10]. It supplies the neutral bilinear current net and the common stress tensor. It does not contain (2.10): a spin/lattice exponential changes sectors and is not a fermion bilinear.

Theorem 3.2 (Energy bounds and smearing for unitary VOAs)**.** *Let V be a unitary VOA with conformal Hamiltonian L_0, and let $a \in V$ be homogeneous. The field $Y(a,z) = \sum_n a_n z^{-n-d_a}$ is polynomially energy bounded if there exist $M > 0$ and integers $s, p \geq 0$ such that*

$$\|a_n b\| \leq M(1 + |n|)^s \, \|(1 + L_0)^p b\| \tag{3.1}$$

for every finite-energy vector b. Under this hypothesis, for every $f \in C^\infty(S^1)$ the series $Y(a,f) = \sum_n \widehat{f}_n a_n$ converges on finite-energy vectors, defines a closable operator, preserves $\mathcal{H}^\infty = \bigcap_{q\geq 0} \mathrm{Dom}(L_0^q)$, and obeys

$$\|Y(a,f)b\| \leq M \, \|f\|_{H^{s+1}} \, \|(1+L_0)^p b\| \,. \tag{3.2}$$

If a unitary VOA is generated by homogeneous energy-bounded fields, all of its fields are energy bounded. A simple unitary VOA generated by weight-one affine currents and conformal vectors is strongly local; its smeared fields generate an irreducible conformal net.

This combines CKLW Proposition 6.1, Theorems 6.8, 8.1 and 8.3 [4], in exactly the direction used below. The newer extension theorem also states that every positive-definite even lattice VOA is energy bounded [5, Proposition 4.2]. These are *continuum* statements; they do not assert a bound uniform in a lattice cutoff.

Theorem 3.3 (Kawahigashi–Longo–Müger)**.** *Let $\mathcal{N}$ be a local irreducible conformal net satisfying the split property and strong additivity, and suppose its two-interval inclusion*

$$\mathcal{N}(E) \subset \widehat{\mathcal{N}}(E) := \mathcal{N}(E')'$$

has finite index. Then $\mathcal{N}$ is completely rational, its DHR category is a nondegenerate finite braided tensor category, and

$$\mu_{\mathcal{N}} = [\widehat{\mathcal{N}}(E) : \mathcal{N}(E)] = \sum_{\rho \in \mathrm{Irr}(\mathcal{N})} d(\rho)^2. \tag{3.3}$$

If $\mathcal{N} \subset \mathcal{M}$ is an irreducible finite-index local extension of completely rational nets, then

$$\mu_{\mathcal{M}} = \frac{\mu_{\mathcal{N}}}{[\mathcal{M} : \mathcal{N}]^2}. \tag{3.4}$$

In particular $\mu_{\mathcal{M}} = 1$ exactly when the vacuum is the only irreducible DHR sector.

This is the multi-interval/global-index and extension theorem of Kawahigashi–Longo–Müger [8].

Theorem 3.4 (Q-systems and local extensions)**.** *Let $N \subset M$ be a finite-index inclusion of factors. A conjugate of the inclusion determines a standard Q-system (θ, w, x) in $\mathrm{End}(N)$, and every standard Q-system reconstructs the inclusion up to isomorphism fixing N. If the endomorphisms are DHR endomorphisms of a conformal net, the reconstructed extension is local if and only if the Q-system is commutative with respect to the DHR braiding. For a finite outer group action $G \curvearrowright M$, $N = M^G$ has index $|G|$, group-average expectation, and the regular Q-system.*

This is the Longo–Rehren reconstruction in the local-extension form [9].

Theorem 3.5 (Trace class implies split)**.** *Let $\mathcal{N}$ be a conformal net with conformal Hamiltonian L_0. If*

$$\mathrm{Tr}(e^{-\beta L_0}) < \infty \qquad \textit{for every } \beta > 0, \tag{3.5}$$

then $\mathcal{N}$ has the split property. More quantitatively, the trace-class condition implies the conformal Buchholz–Wichmann nuclearity maps are nuclear; the Buchholz–Wichmann condition implies statistical independence for strictly separated intervals.

The implication is due to Buchholz–Wichmann and its conformal refinements by Buchholz–D'Antoni–Longo [2, 1].

Theorem 3.6 (Central charge eight and the E_8 theory)**.** *Let V be a simple, rational, C_2-cofinite, holomorphic unitary VOA of CFT type and central charge* 8*. Then its character is*

$$\mathrm{ch}_V(\tau) = \frac{E_4(\tau)}{\eta(\tau)^8} = q^{-1/3}(1 + 248q + 4124q^2 + 34752q^3 + \cdots), \tag{3.6}$$

V_1 is the simple Lie algebra E_8 at level one, and $V \cong V_{E_8}$. The strongly local net associated by the CKLW construction is the level-one E_8 current net.

The VOA classification statement is the central-charge-eight theorem of Dong–Mason [6]; the last sentence is CKLW. We stress that KLM alone proves holomorphy, not the name "E_8". The VOA reconstruction/weight-one argument is the additional identification step.

4 L1: uniform energy bounds

4.1 The continuum estimate is complete

Lemma 4.1 (Linear energy bound for the limiting norm-two field)**.** *In the level-one lattice VOA of the glued rank-eight lattice, the norm-two vector e^λ is a weight-one affine current. For every $f \in C^\infty(S^1)$ there is a constant c_f such that*

$$\left\| Y(e^\lambda, f)\xi \right\| \le c_f \left\| (1 + L_0)\xi \right\|. \tag{4.1}$$

Proof. The norm formula gives $h_\lambda = \|\lambda\|^2/2 = 1$. The weight-one subspace of the lattice VOA is the direct sum of the Cartan currents and the root vectors e^α for $\|\alpha\|^2 = 2$. Their modes satisfy the level-one affine commutation relations. Positivity of the Sugawara expression for L_0 gives, for a fixed normalized current J,

$$\|J_n\xi\| \le C(1 + |n|) \left\| (1 + L_0)^{1/2}\xi \right\|. \tag{4.2}$$

Indeed, on each L_0 eigenspace, the positive quadratic Sugawara sum contains $J_{-n}J_n$; commuting a negative mode to the right contributes the central term $|n|$. Summing $\sum_n |\widehat{f}_n| \, \|J_n\xi\|$ and applying Cauchy–Schwarz with $\sum_n (1 + |n|)^4 |\widehat{f}_n|^2 < \infty$ gives

$$\|J(f)\xi\| \le C \, \|f\|_{H^2} \, \|(1 + L_0)\xi\|.$$

This is also the weight-one case of Theorem 3.2. ☐

4.2 The edge-projected lattice Green form

We now define and estimate the concrete G_N used by the collar bosonization. It is important that this is a definition from the frozen QWZ edge band, not a continuum ansatz.

Use antiperiodic (NS) momenta

$$p_n = \frac{2\pi}{N}\left(n + \frac{1}{2}\right), \qquad -\frac{N}{2} \le n < \frac{N}{2}, \tag{4.3}$$

and Hardy occupation $\nu_n = \mathbf{1}_{\{n<0\}}$. The same argument works after a fixed deck shift because only the number of indices crossing the Hardy cut is used. Let $u(p)$ be the normalized top-edge eigenvector of the $2N_y$-dimensional strip matrix through $p = 0$. Away from the crossing choose any normalized continuation; the global bound below uses no regularity there.

For $1 \le m \le N/2$, define the finite edge-current structure factor

$$S_N(m) := \sum_{n=-m}^{-1} |\langle u(p_{n+m}), u(p_n) \rangle|^2, \tag{4.4}$$

and extend it evenly to negative m. With the lattice derivative (2.9), the Green multiplier is

$$g_N(m) := \frac{S_N(m)}{2\omega_N(m)^2}. \tag{4.5}$$

Let $\mathbb{M}_N = \{-N/2+1, \dots, -1, 1, \dots, N/2\}$ be the fundamental nonzero Fourier representatives. For an eight-component zero-mean charge q in the band-limited trigonometric space $\mathcal{V}_N$, put

$$\mathcal{Q}_N(q) := \langle q, G_N q\rangle = \sum_{a=1}^{8} \sum_{m\in\mathbb{M}_N} g_N(m)|\widehat{q}_a(m)|^2. \tag{4.6}$$

The continuum chiral form is

$$\mathcal{Q}_{\mathrm{ch}}(q) = \frac{1}{2}\sum_{a=1}^{8}\sum_{m\neq 0} \frac{|\widehat{q}_a(m)|^2}{|m|}. \tag{4.7}$$

The diagonal logarithm is omitted on both sides by the same Wick normal ordering.

Lemma 4.2 (Exact transverse edge doublet)**.** *At $p = 0$, the $M = 1$, $N_y = 8$ strip Hamiltonian has two zero modes, one on each boundary. The remaining spectrum is $\{+1, -1\}$, each with multiplicity seven. The two eigenbranches through zero are analytic near $p = 0$, have slopes $+1$ and -1, and are separated there from the bulk complement by $\delta_{\mathrm{bulk}} = 1$.*

Proof. At $p = 0$ the onsite term in (2.1) vanishes. The transverse hopping is

$$T_y = \frac{\sigma_y}{2i} - \frac{\sigma_z}{2}.$$

It has rank one and singular value one. In the eigenbasis of σ_x, the open strip is the orthogonal sum of seven two-site blocks $\left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$ and two unpaired boundary vectors. Hence the stated spectrum and $\delta_{\mathrm{bulk}} = 1$. The derivative of the onsite matrix at zero is σ_x; its compression to the two boundary zero modes is diagonal with eigenvalues $+1$ and -1. Analytic perturbation theory for this isolated two-dimensional spectral subspace, followed by diagonalization of the first derivative inside it, gives two analytic branches with those slopes. □

Lemma 4.3 (Sharp lattice-symbol comparison)**.** *For every N and $0 < |m| \le N/2$,*

$$\frac{2}{\pi}|m| \le \omega_N(m) \le |m|. \tag{4.8}$$

Both constants are sharp.

Proof. Set $x = \pi|m|/N \in (0, \pi/2]$. Concavity of $\sin x$ on this interval places its graph above the chord joining $(0, 0)$ and $(\pi/2, 1)$, so $\sin x \ge 2x/\pi$. The inequality $\sin x \le x$ gives the upper bound. The lower constant is attained at $|m| = N/2$, and the upper constant is approached as $|m|/N \to 0$. □

Lemma 4.4 (Particle–hole Schur bound)**.** *For every N and $0 < |m| \le N/2$,*

$$0 \le S_N(m) \le |m|. \tag{4.9}$$

Proof. For $m > 0$, exactly m occupied Hardy indices $n = -m, \dots, -1$ are carried across the Fermi cut. Every summand in (4.4) lies in $[0, 1]$ by Cauchy–Schwarz and the normalization of $u(p)$. This proves the claim; evenness handles $m < 0$. Equivalently, this is the Schur bound for the compression of the momentum-shift partial isometry between occupied and unoccupied edge subspaces. □

Lemma 4.5 (Bulk resolvent comparison)**.** *Assume the bulk-edge decomposition* (2.4)*. For z on a contour at distance $\delta_{\mathrm{bulk}}/2$ from the bulk spectrum and for two bulk Hamiltonians $H_{\mathrm{b}}, \widetilde{H}_{\mathrm{b}}$,*

$$\left\|(z-H_{\mathrm{b}})^{-1}-(z-\widetilde{H}_{\mathrm{b}})^{-1}\right\| \le \frac{4}{\delta_{\mathrm{bulk}}^2}\left\|H_{\mathrm{b}}-\widetilde{H}_{\mathrm{b}}\right\|. \tag{4.10}$$

Consequently the corresponding bulk spectral projections differ by at most a contour-length constant times the right-hand side.

Proof. The resolvent identity gives

$$(z-H_{\mathrm{b}})^{-1}-(z-\widetilde{H}_{\mathrm{b}})^{-1} = (z-H_{\mathrm{b}})^{-1}(H_{\mathrm{b}}-\widetilde{H}_{\mathrm{b}})(z-\widetilde{H}_{\mathrm{b}})^{-1}.$$

Both resolvent norms are at most $2/\delta_{\mathrm{bulk}}$. Integrating the identity in the Riesz formula proves the projection statement. □

Remark 4.6. Lemma 4.5 controls the massive transverse complement. It supplies the isolated analytic edge subspace used in Lemma 4.2, but it cannot by itself furnish the $|m|^{-1}$ bound in (4.7). That bound comes from the exact Hardy particle–hole count and Lemma 4.3.

Theorem 4.7 (Uniform Green-form domination and Sobolev convergence)**.** *For the edge-projected frozen $M = 1$, $N_y = 8$ QWZ collar, every zero-mean $q \in \mathcal{V}_N \otimes \mathbb{C}^8$ satisfies*

$$0 \le \mathcal{Q}_N(q) \le \frac{\pi^2}{4}\mathcal{Q}_{\mathrm{ch}}(q). \tag{4.11}$$

Thus one may take the all-volume constant

$$K_G = \frac{\pi^2}{4} = 2.467401100272\ldots. \tag{4.12}$$

If P_N is Fourier projection onto $|m| \le N/2$, then $G_N P_N \to G_{\mathrm{ch}}$ in local Sobolev quadratic-form norm:

$$|\mathcal{Q}_N(P_N q) - \mathcal{Q}_{\mathrm{ch}}(q)| \le \varepsilon_N \|q\|^2_{H^2(S^1;\mathbb{C}^8)}, \qquad \varepsilon_N \longrightarrow 0. \tag{4.13}$$

Proof. Lemmas 4.3 and 4.4 give, mode by mode,

$$0 \le g_N(m) = \frac{S_N(m)}{2\omega_N(m)^2} \le \frac{|m|}{2(2|m|/\pi)^2} = \frac{\pi^2}{8|m|} = \frac{\pi^2}{4}\frac{1}{2|m|}. \tag{4.14}$$

Multiplication by $|\widehat{q}_a(m)|^2$ and summation over the modes and eight independent components proves (4.11). Notice that the number of components does not multiply K_G.

It remains to prove convergence. Fix m. The $|m|$ terms in (4.4) involve momenta satisfying $|p_n| + |p_{n+m}| = O_m(N^{-1})$. By Lemma 4.2, $u(p)$ is norm-continuous there, uniformly over this finite set. Consequently

$$S_N(m) \longrightarrow |m|, \qquad \frac{\omega_N(m)}{|m|} \longrightarrow 1, \qquad g_N(m) \longrightarrow \frac{1}{2|m|}. \tag{4.15}$$

Define

$$\varepsilon_N := \sup_{0<|m|\le N/2} \frac{|g_N(m) - (2|m|)^{-1}|}{(1+m^2)^2} + \sup_{|m|>N/2} \frac{1}{2|m|(1+m^2)^2}. \tag{4.16}$$

For any fixed M, the first supremum over $|m| \le M$ tends to zero by (4.15). For $|m| > M$, (4.14) bounds the numerator by $(K_G + 1)/(2|m|)$, uniformly in N; hence the remaining supremum tends to zero as $M \to \infty$. Thus $\varepsilon_N \to 0$. Parseval's identity and the definition of the H^2 norm now give (4.13). □

Theorem 4.8 (L1 after the proved Green estimate)**.** *For the edge-projected bosonization, assume only that the normalized cocycles and zero-mode charge shifts have norm one in all four grades. Then for every smooth f the bound (1.1) holds with $p = 1$ and*

$$C_f \le C(\pi^2/4, \lambda)\, \|f\|_{H^2}\,, \tag{4.17}$$

independent of N.

Proof. Bosonic Wick ordering factors a mode of (2.10) into the norm-one zero-mode shift and exponentials of creation and annihilation currents. The normal-ordered lattice-VOA estimate is the finite-mode version of Theorem 3.2; its constant depends on the Gaussian contraction only through the Green quadratic form. Theorem 4.7 bounds that contraction uniformly by the continuum form with $K_G = \pi^2/4$. The norm-two vector has weight one, so Lemma 4.1 and (3.2) give degree $p = 1$ and (4.17). Wick scalar factors are bounded by the same quadratic-form domination. □

Remark 4.9 (Scope of the proved UGF)*.* Theorem 4.7 is an all-N theorem for the edge-projected bosonization used in (2.10); it is not a fit to the six tested sizes. It does not prove that every microscopic full-CAR density operator belongs to this bosonic edge algebra. Excluding additional bulk/ultraviolet local scaling operators is exactly the separate (ALG-EXH) condition. A completely gapless transverse covariance would contribute $O(N)$ particle–hole channels already at $m = 1$ and would violate (4.11); the isolated one-branch hypothesis is therefore load-bearing.

Finite audit (numerical support, not proof). The companion probe `experiments/tfpt-discovery/mmst_ugf_bound_probe.py` checks every displayed inequality at $N = 16, 24, 32, 48, 64, 96$. It finds $\max_{N,m} 2|m|g_N(m) = 1.029399027646 < K_G$, verifies $\delta_{\text{bulk}} = 1$ and the unit edge slope, and rejects both a gapless transverse mutant (whose ratio grows from 64.829 to 384.137) and the false symbol constant $\omega_N(m) \ge |m|$.

5 L2: strong convergence and local algebras

5.1 Explicit dyadic mode and Fock isometries

Let

$$\mathcal{K}_N = \ell^2(\{m \in \mathbb{Z} : 0 < |m| \le N/2\}) \otimes \mathbb{C}^8 \tag{5.1}$$

with the single Nyquist representative convention of $\mathbb{M}_N$. Define

$$j_{N,2N}(e_m \otimes e_a) = e_m \otimes e_a, \qquad m \in \mathbb{M}_N, \quad 1 \le a \le 8. \tag{5.2}$$

Thus $j_{N,2N}$ is the zero-extension/Shannon embedding of the old Fourier modes into the refined lattice. It is an isometry and

$$j_{2N,4N}\, j_{N,2N} = j_{N,4N}. \tag{5.3}$$

On the charge-lattice sum of symmetric bosonic Fock spaces set

$$V_{N,2N} := \mathbf{1}_{\ell^2(L_{\text{charge}})} \otimes \Gamma_s(j_{N,2N}). \tag{5.4}$$

Second quantization preserves isometries and composition, so $V_{N,2N}^* V_{N,2N} = 1$ and $V_{2N,4N} V_{N,2N} = V_{N,4N}$. The charge-lattice identity makes the same statement valid in every one of the four λ grades.

5.2 The multiplier rate extracted from UGF

For $m > 0$ define the vertex multiplier

$$r_N(m) := \frac{\sqrt{mS_N(m)}}{\omega_N(m)} = \sqrt{2m\, g_N(m)} \quad (m \le N/2), \qquad r_N(m) := 0 \quad (m > N/2). \tag{5.5}$$

The creation part of the finite vertex exponential contains $r_N(m)a_{-m}/m$; the annihilation part is its adjoint. Theorem 4.7 gives $0 \le r_N(m) \le \pi/2$.

Lemma 5.1 (Uniform quadratic multiplier error)**.** *For all even $N \ge 16$ and $1 \le m \le N/2$,*

$$|r_N(m) - 1| \le A_0 \frac{m^2}{N^2}, \qquad A_0 := 4096 \left(1 + \frac{\pi}{2}\right). \tag{5.6}$$

Consequently, for $1 \le m \le N$,

$$|r_{2N}(m) - r_N(m)| \le A_{\mathrm{TEL}} \frac{m^2}{N^2}, \qquad A_{\mathrm{TEL}} := \frac{5}{4} A_0 = 5120 \left(1 + \frac{\pi}{2}\right). \tag{5.7}$$

Proof. Put $t = m/N$. For $t \le 1/64$, every momentum entering (4.4) lies in $[-\pi/32, \pi/32]$. Differentiating the explicit $M = 1$ transfer recurrence gives

$$\sup_{|p| \le \pi/32} \|u'(p)\| \le 4. \tag{5.8}$$

For completeness, the decaying transfer root is $\rho(p) = 1 - \cos p$; on this interval $|\rho| \le 1 - \cos(\pi/32)$ and $|\rho'| \le \sin(\pi/32)$. Differentiating the normalized geometric vector gives a norm below 1, while the opposite-boundary correction is bounded by $2|\rho|^{N_y}/(1 - |\rho|)$ and its derivative by the corresponding geometric derivative; their sum is below 4 for $N_y = 8$. Therefore

$$0 \le 1 - \frac{S_N(m)}{m} \le \|u(p_{n+m}) - u(p_n)\|^2 \le 64\pi^2 t^2. \tag{5.9}$$

Also, elementary Taylor remainder and Lemma 4.3 give

$$0 \le \frac{\pi t}{\sin(\pi t)} - 1 \le \pi^2 t^2 \qquad (0 \le t \le 1/2). \tag{5.10}$$

Since $r_N = (\pi t / \sin \pi t)\sqrt{S_N/m}$, these estimates prove (5.6) for $t \le 1/64$ with a constant much smaller than A_0. For $t \ge 1/64$, the UGF bound gives $|r_N - 1| \le 1 + \pi/2 \le A_0 t^2$, proving the global statement.

For $m \le N/2$, apply (5.6) at N and $2N$ and use the triangle inequality, giving the factor $1 + 1/4 = 5/4$. For $N/2 < m \le N$, $r_N(m) = 0$ and $r_{2N}(m) \le \pi/2 \le 2\pi m^2/N^2 < A_{\mathrm{TEL}} m^2/N^2$. □

Lemma 5.2 (Lipschitz Wick estimate)**.** *Let r, s be two cutoff multiplier sequences bounded by $\pi/2$ and suppose $|r(m) - s(m)| \le am^2$. For the norm-two charge λ and every $f \in C^\infty(S^1)$,*

$$\left\| (Y_r(e^\lambda, f) - Y_s(e^\lambda, f))(1 + L_0)^{-3} \right\| \le C_\lambda\, a\, \|f\|_{H^4}, \tag{5.11}$$

where C_λ is independent of both cutoffs. The same estimate holds for the adjoints.

Proof. Interpolate $r_\tau = (1 - \tau)r + \tau s$ and differentiate the finite normal-ordered exponential. Every differentiated monomial has exactly one insertion

$$J_{s-r}(z) = \sum_{m>0} \frac{s(m) - r(m)}{m} (a_{-m} z^m - a_m z^{-m})$$

and otherwise the vertex exponential with multiplier r_τ. The oscillator inequalities $\left\| a_{\pm m}(1+L_0)^{-1/2} \right\| \le \sqrt{m+1}$, followed by Cauchy–Schwarz in m and four integrations by parts in the smearing variable, give

$$\left\| :J_{s-r} Y_{r_\tau}(e^\lambda): (f)(1+L_0)^{-3} \right\| \le C_\lambda a \left\| f \right\|_{H^4}.$$

The constant is uniform because $0 \le r_\tau(m) \le \pi/2$, so all undifferentiated Wick contractions are dominated by Theorem 4.7. Integrating over $\tau \in [0,1]$ proves (5.11). The finite-cutoff calculation is an absolutely convergent sum after the displayed graph weight; monotone cutoff removal gives the asserted bound. Replacing λ by $-\lambda$ and conjugating the coefficients proves the adjoint statement. □

Theorem 5.3 (TEL-a: dyadic charged-field graph defect). *The explicit isometries* (5.4) *satisfy, for every $f \in C^\infty(S^1)$,*

$$\begin{aligned} &\left\| (\Psi_{\lambda,2N}(f) V_{N,2N} - V_{N,2N}\Psi_{\lambda,N}(f))(1+L_{0,N})^{-3} \right\| \\ &\quad + \left\| (\Psi_{\lambda,2N}(\overline{f})^* V_{N,2N} - V_{N,2N}\Psi_{\lambda,N}(\overline{f})^*)(1+L_{0,N})^{-3} \right\| \\ &\qquad \le 2C_\lambda A_{\mathrm{TEL}} \left\| f \right\|_{H^4} N^{-2}. \end{aligned} \tag{5.12}$$

Thus TEL-a holds with $p = 3$ and $\epsilon = 1$.

Proof. After applying $V_{N,2N}$, the coarse field is the fine Fock-space vertex field with multiplier r_N extended by zero on the new modes. Apply Lemma 5.2 with $a = A_{\mathrm{TEL}} N^{-2}$ from Lemma 5.1; apply it again to the adjoint. This proves (5.12). □

Lemma 5.4 (Summable graph defects imply strong convergence). *The dyadic estimate* (5.12) *implies that the transported fields and adjoints converge on $\mathcal{D}_{\mathrm{fin}}$ along every dyadic cofinal sequence.*

Proof. Let ξ have energy at most E and represent it at scale N_0. For $K > L$, insert the compatible isometries and telescope:

$$\left\| \Psi^{\mathrm{tr}}_{\lambda,2^K N_0}(f)\xi - \Psi^{\mathrm{tr}}_{\lambda,2^L N_0}(f)\xi \right\| \le 2C_\lambda A_{\mathrm{TEL}} \left\| f \right\|_{H^4} (1+E)^3 \sum_{j=L}^{K-1} (2^j N_0)^{-2}. \tag{5.13}$$

The geometric tail tends to zero. The adjoint half of (5.12) gives graph convergence, and L1 makes the limit closable. A dyadic subsystem is cofinal in the scaling parameter, so this is sufficient for the edge-mode scaling limit. □

Finite audit (numerical support, not proof). The companion probe `experiments/tfpt-discovery/mmst_tel_probe.py` checks (5.2)–(5.7) at $N = 16, 24, 32, 48, 64, 96$. The Gram and compatibility errors are exactly zero, the measured coefficient envelope is $\max N^2 |r_{2N}(m) - r_N(m)|/m^2 = 3.420384012310$, and a smooth-test Wick seminorm fits $N^{-2.022861}$. A wrong-rate $1/N$ mutant and a non-isometric embedding mutant are both rejected.

Remark 5.5 (Local support). The Shannon embedding (5.2) is exact in momentum and is not sharply support-preserving in position. Theorem 5.3 therefore proves graph convergence of the edge-projected fields, not by itself convergence of sharply localized microscopic algebras. The support/locality statement remains explicitly in (ALG-EXH).

5.3 TEL-b: the two-edge full-CAR remainder

The exact Hardy model gives the quarter-twist coefficient

$$\left\|P_0 - P_{1/4}\right\|_{\mathrm{HS}}^2 = \frac{2}{\pi^2}\log N + O(1), \qquad \frac{2}{\pi^2} = 0.202642367\ldots. \tag{5.14}$$

For the frozen QWZ exhaustion, the computed decomposition is

$$C_N^{(0)} - C_N^{(1)} = D_{N,\mathrm{top}}^{(1/4)} \oplus D_{N,\mathrm{bottom}}^{(1/4)} + R_N, \tag{5.15}$$

with

$$\max_{N\in\{16,24,32,48,64,96\}} \|R_N\|_{\mathrm{HS}} = 2.105079358 < 2.11. \tag{5.16}$$

A $1/N$ fit predicts 2.107565497, with maximum fit residual 2.349×10^{-5}. These values support boundedness of R_N but do not prove either

$$\sup_N \|R_N\|_{\mathrm{HS}} \le 2.11 \tag{5.17}$$

for the full-CAR remainder.

Proposition 5.6 (Exact Hankel reduction of TEL-b)**.** *Let*

$$C(p) = \mathbf{1}_{(-\infty,0)}(h_{\mathrm{strip}}(p)) + \frac{1}{2}\mathbf{1}_{\{0\}}(h_{\mathrm{strip}}(p))$$

be the explicit 16×16 *frozen QWZ Fermi symbol and let* $P_{\mathrm{t}}, P_{\mathrm{b}}$ *be the rank-one boundary projections at* $p = 0$*. The symbol left after the constant-profile Hardy subtraction is*

$$B(p) = C(p) - \mathbf{1}_{(0,\pi)}(p)P_{\mathrm{t}} - \mathbf{1}_{(-\pi,0)}(p)P_{\mathrm{b}}. \tag{5.18}$$

Writing $B(p) = \sum_{k\in\mathbb{Z}} B_k e^{ikp}$*, every finite Hankel corner obeys*

$$\|H_N(B)\|_{\mathrm{HS}}^2 \le \sum_{k\neq 0} |k| \, \|B_k\|_{\mathrm{F}}^2 \,. \tag{5.19}$$

Hence an all-N *TEL-b proof by this subtraction would follow from*

$$16 \sum_{k\neq 0} |k| \, \|B_k\|_{\mathrm{F}}^2 \le C_R^2. \tag{5.20}$$

Proof. In the untwisted Hardy basis, multiplication by B has matrix entries B_{j-k}. Its off-diagonal Hardy corner has entries B_{j+k+1}. Counting the pairs $(j,k) \in \mathbb{N}_0^2$ with $j+k+1 = \ell$ gives the exact Hilbert–Schmidt identity

$$\|H(B)\|_{\mathrm{HS}}^2 = \sum_{\ell\geq 1} \ell \, \|B_\ell\|_{\mathrm{F}}^2 \,.$$

Finite corners are compressions and cannot increase this norm. The opposite edge uses the negative coefficients. There are at most four corner/orientation contributions; the triangle inequality gives the safe factor sixteen after squaring in (5.20). This is the $p = 2$ case of the Peller $B_{2,2}^{1/2}$ criterion [11]. □

Remark 5.7 (Second-angle verdict: `TEL-B-EXTERNAL`)*.* The new Hankel route does not certify the requested constant. On a 4096-point symbol grid the companion probe finds

$$\sum_{k\neq 0} |k| \, \|B_k\|_{\mathrm{F}}^2 = 4.482956087, \qquad 4\,\|B\|_{\dot H^{1/2}} = 8.469196974 > 3.$$

The constant edge-profile subtraction cancels the infrared jump at $p = 0$ but exposes a Brillouin-seam ultraviolet contribution; the raw symbol seminorm on the same grid is 2.570232705. Thus neither a $C_R = 2.5$ nor a $C_R = 3$ proof follows, and the value is numerical rather than an all-mode enclosure. UGF cannot repair this because it controls only the scalar edge branch, whereas (5.18) is a matrix-valued transverse cancellation. This was the second distinct in-house attempt; by the stop rule TEL-b is now typed `TEL-B-EXTERNAL`. No all-N constant is claimed.

Remark 5.8 (Round-3 supersession, 2026-09-05)*.* The preceding verdict records the historical raw-Hankel attempt. It is not the current TEL-b residue. Native 256-bit Arb/Acb arithmetic on 5812 rational cells now proves

$$\mathrm{TV}_p(F'') \le 15.8176959629889243576\ldots < 15.928674, \qquad K_3^{\mathrm{TV}} < 2.517464500835,$$

including the exact Hilbert–Schmidt cut jump 2 (`v1022_telb_round3_certificates.py`). Separately, a 100-bit certificate over 3888 analytic rectangles proves $C_\star < 0.1525$, and the Peano/Euler argument proves $\sup_k |\varphi_N(k)| < 0.509/N$ for every even $N \ge 16$ (`v1025_telb_c1_c2a_certificates.py`). These close the TV, C1 and C2a inputs. Round 4 additionally proves $\mathrm{CF} < 0.065697$, $\mathrm{DG} < 0.042959$, and the complete relaxed norm $\|R_N\|_{\mathrm{HS}} < 2.995906 < 3$ for every even $N \ge 16$, including the finite-grid aliases (`v1026_telb_round4_closure.py`). The remaining microscopic algebra identification is not a consequence of this norm theorem.

5.4 Third angle: certified finite part plus a dyadic tail

There are two natural full-CAR one-particle embeddings from circumference N to $2N$. The low-energy embedding J_N^{low} preserves the integer Fourier label n, while the physical-momentum dilation J_N^{dil} sends $n \mapsto 2n$. The proposed tail mechanism would require, for one of them,

$$\|R_{2N} - J_N R_N J_N^*\|_{\mathrm{HS}} \le D N^{-2} \tag{5.21}$$

with one explicit D. The reverse triangle inequality would then give

$$\big|\|R_{2N}\|_{\mathrm{HS}} - \|R_N\|_{\mathrm{HS}}\big| \le D N^{-2}, \tag{5.22}$$

and the dyadic tail from N_0 would be at most $D N_0^{-2}/(1 - 1/4)$.

Proposition 5.9 (The TEL-a embedding does not supply the proposed remainder tail)**.** *On the frozen finite matrices, neither natural embedding exhibits the defect required by* (5.21)*. The measured values are*

$N \to 2N$	$\left\|R_{2N} - J_N^{\mathrm{low}} R_N (J_N^{\mathrm{low}})^*\right\|_{\mathrm{HS}}$	$\left\|R_{2N} - J_N^{\mathrm{dil}} R_N (J_N^{\mathrm{dil}})^*\right\|_{\mathrm{HS}}$
$16 \to 32$	2.071346866	2.066044074
$32 \to 64$	2.067361892	2.072036965
$64 \to 128$	2.065611659	2.075227063
$96 \to 192$	2.065062404	2.076335188

(5.23)

Thus the certified-tail argument cannot inherit the proved TEL-a N^{-2} comparison.

Proof. In the untwisted Fourier basis, construct R_N exactly as in (5.15); both embeddings are coordinate isometries. For either J,

$$\|R_{2N} - J R_N J^*\|_{\mathrm{HS}}^2 = \|R_{2N}\|_{\mathrm{HS}}^2 + \|R_N\|_{\mathrm{HS}}^2 - 2\Re \langle J^* R_{2N} J, R_N\rangle_{\mathrm{HS}}\,.$$

The companion probe evaluates this identity without forming J and reproduces the established finite norms. The table is numerical evidence, not an interval theorem; it shows that the proposed small TEL-a-type defect is absent on the tested scales. Therefore high-precision recalculation of the same finite values cannot certify the missing infinite tail. ☐

Remark 5.10 (The strictly smaller dyadic residue). The scalar norm increments are instead

$$0.007486497094, \quad 0.003790483818, \quad 0.001913049365, \quad 0.001279839965$$

for the four rows of (5.23); the products N times these increments are

$$0.119783953, \quad 0.121295482, \quad 0.122435159, \quad 0.122864637.$$

This suggests, but does not prove,

$$\big|\|R_{2N}\|_{\mathrm{HS}} - \|R_N\|_{\mathrm{HS}}\big| \le \frac{A_R}{N} \qquad (N \ge 96). \tag{5.24}$$

Unlike an adjacent-size $1/N$ estimate, (5.24) is summable on a dyadic tower. If proved, it gives

$$\sup_{k\ge 0} \|R_{96\cdot 2^k}\|_{\mathrm{HS}} \le \|R_{96}\|_{\mathrm{HS}} + \frac{2A_R}{96}. \tag{5.25}$$

The edge-mode scaling theorem uses a cofinal dyadic subsystem, so (5.25), not a supremum over every integer N, is the minimal required statement. The six reported finite values are ordinary double-precision diagnostics. Arbitrary-precision `mpmath` output alone is not a rigorous interval certificate; because the prerequisite tail theorem failed, no floating-point value is relabelled as certified.

Remark 5.11 (The dyadic-increment route is retired). Equations (5.24)–(5.25) are retained as an audit trail, not as the current proof route. The cover/sawtooth reduction and the round-3 C1/C2a/TV certificates supersede this proposed scalar increment. No all-N conclusion is imported from the measurements above.

Third-angle audit. `experiments/tfpt-discovery/mmst_telb_tail_probe.py` reproduces all six established finite norms, evaluates both embeddings through $N = 192$, verifies the reverse-triangle inequality, and rejects an N^{-1} mutant as an N^{-2} tail. Its verdict is `TAIL-TELESCOPING-FAILS`; all diagnostic and honesty guards pass.

5.5 Relative locality of the charged and neutral fields

Theorem 5.12 (Relative locality survives the graph limit)**.** *The limiting edge-projected charged field is relatively local with respect to the neutral $D_5 \oplus A_3$ current net.*

Proof. For every neutral root α, the frozen generator satisfies

$$\langle \lambda, \alpha \rangle \in \mathbb{Z}. \tag{5.26}$$

On the displayed simple-root basis the eight pairings are $[0, 0, 0, 0, 1, 1, 0, 0]$; integrality for the full root lattice follows by linearity. Therefore the finite vertex exchange phase is $e^{2\pi i \langle \lambda, \alpha \rangle} = 1$. If f and g have disjoint supports, the finite collar consequently satisfies

$$[\Psi_{\lambda,N}(f), W_{\alpha,N}(g)]\xi = 0$$

on the finite-energy core, where $W_{\alpha,N}(g)$ is a bounded neutral Weyl generator.

Transport to the common Hilbert space with (5.4). Theorem 5.3 gives graph convergence of the charged field and its adjoint; Osborne–Stottmeister gives strong convergence of the bounded neutral Weyl operators. For a common-core vector ξ,

$$\begin{aligned} W_\alpha(g)\Psi_\lambda(f)\xi &= \lim_N W_{\alpha,N}(g)\Psi_{\lambda,N}(f)\xi \\ &= \lim_N \Psi_{\lambda,N}(f)W_{\alpha,N}(g)\xi = \Psi_\lambda(f)W_\alpha(g)\xi. \end{aligned}$$

The uniform graph bound justifies both products. Closability extends the equality from the core, and the Weyl generators strongly generate the neutral local algebra. □

5.6 Outerness of the limiting four-grade action

Theorem 5.13 (Outerness from the connected simple-current Q-system)**.** *On every limiting local factor generated by the neutral algebra and the four charged grades, every nonidentity element of the $\mathbb{Z}_4$ gauge action is outer.*

Proof. Let $\iota : \mathcal{A}(I) \hookrightarrow \mathcal{B}^{\mathrm{gen}}(I)$ be the local inclusion. Its canonical endomorphism has the multiplicity-free decomposition

$$\bar{\iota}\iota \simeq \mathbf{1} \oplus \rho_\lambda \oplus \rho_{2\lambda} \oplus \rho_{3\lambda}. \tag{5.27}$$

The four sectors are inequivalent: their exact grades are distinct and $[\lambda]$ has order four. Only the first summand is the vacuum, so

$$\mathcal{A}(I)' \cap \mathcal{B}^{\mathrm{gen}}(I) \cong \mathrm{Hom}(\mathbf{1}, \bar{\iota}\iota) = \mathbb{C}. \tag{5.28}$$

Suppose $\gamma_g = \mathrm{Ad}\, u$ for a nontrivial $g \in \mathbb{Z}_4$ and $u \in \mathcal{B}^{\mathrm{gen}}(I)$. Since γ_g fixes $\mathcal{A}(I)$ pointwise, $u \in \mathcal{A}(I)' \cap \mathcal{B}^{\mathrm{gen}}(I) = \mathbb{C}$, so $\gamma_g = \mathbf{1}$. This contradicts faithfulness: on the grade-one field it acts by the nontrivial character $\gamma_g(\Psi_\lambda) = \chi_1(g)\Psi_\lambda$. Hence γ_g is outer.

There is no finite-dimensional outerness witness: every automorphism of M_{256} is inner, and $U = \mathrm{diag}(1_{64}, i1_{64}, -1_{64}, -i1_{64})$ is a charge-zero implementing unitary. Outerness is a local-factor limit statement derived from (5.28), not inherited from finite matrix algebras. □

5.7 From fields to local algebras

For an interval I, define the field-generated finite algebra

$$\mathcal{B}_N^{\mathrm{gen}}(I) = W^*\Big(\mathcal{A}_N(I),\ \overline{\Psi_{k\lambda,N}(f)} : k \in \mathbb{Z}_4,\ \mathrm{supp}\, f \subset I\Big). \tag{5.29}$$

Define $\mathcal{B}^{\mathrm{gen}}(I)$ similarly from the limiting fields.

Lemma 5.14 (Graph convergence of a complete generating family)**.** *Assume L1, the graph convergence* (2.12) *for a support-preserving countable dense family of test functions, strong locality of the limiting fields, and the exhaustion condition* (ALG-EXH) *from the final box. Then*

$$\mathcal{B}_N^{\mathrm{gen}}(I) \longrightarrow \mathcal{B}^{\mathrm{gen}}(I) \tag{5.30}$$

in the standard scaling-algebra sense: bounded functions of the generating fields converge strongly, the strong lower and upper limits coincide, and no additional local scaling operator remains.

Proof. Graph convergence of a closed operator and its adjoint implies strong resolvent convergence for its hermitian real and imaginary parts on the common invariant core. Bounded continuous functions of those parts therefore converge strongly. Their spectral projections generate the same von Neumann algebra as the affiliated smeared fields. Thus every element of the limiting generated algebra belongs to the strong lower limit. Conversely, (ALG-EXH) says exactly that every bounded local scaling sequence is approximable by bounded words in this generating family; its weak cluster points lie in $\mathcal{B}^{\mathrm{gen}}(I)$. Hence the upper limit is contained in the lower limit. □

Proposition 5.15 (Sharp Buchholz–Verch exhaustion reduction)**.** *Let $\underline{\mathcal{B}}^{\mathrm{gen}}(I)$ denote the scaling subalgebra generated by the neutral Weyl operators and the four graph-convergent charged grades. Equality with the full microscopic Buchholz–Verch scaling algebra follows if the following two estimates hold:*

$$\begin{gathered}\lim_{E\to\infty} \sup_{N\geq 16} \sup_{\substack{x\in\mathcal{B}_N(I)\\ \|x\|\leq 1}} \Big\|(1 - P_{E,N}^{\mathrm{gen}})e^{-\beta L_{0,N}} x\Omega_N\Big\| = 0 \quad (\beta > 0),\\ \lim_{N\to\infty} \inf_{y_N\in\mathcal{B}_N^{\mathrm{gen}}(I)} (\|(x_N - y_N)P_{E,N}\| + \|(x_N^* - y_N^*)P_{E,N}\|) = 0\end{gathered} \tag{5.31}$$

for every finite E and every norm-bounded, translation-continuous local scaling sequence (x_N).

Proof. The first line is uniform phase-space tightness: after Euclidean energy damping, vectors created by the microscopic unit ball are uniformly approximated by the finite-energy neutral-plus-four-grade subspace. The second line says that no bounded operator sequence has additional matrix elements on any finite-energy sector, for the operator and its adjoint. It gives equality of the strong lower limits on the common finite-energy core. Phase-space tightness then excludes a weak cluster point supported only at energies escaping to infinity, giving equality of the upper limit as well. This is the standard scaling-algebra compactness mechanism of Buchholz–Verch [3], written here as the two concrete estimates the collar must satisfy. □

Remark 5.16 (Verified and missing Buchholz–Verch hypotheses). The frozen lattice system has isotony, exact finite-range locality, uniform norm bounds for scaling families, discrete translation covariance, and a translation-covariant neutral scaling limit. Theorems 4.7, 5.3 and 5.12 add uniform charged energy control, graph convergence and the lower inclusion $\underline{\mathcal{B}}^{\mathrm{gen}}(I) \subset \underline{\mathcal{B}}(I)$. What is not proved is the uniform microscopic phase-space estimate and the second, upper-inclusion line of (5.31). These are precisely the "no extra ultraviolet operators" content; no arithmetic or finite sector census can establish them.

Remark 5.17 (Binding ALG2 correction, 2026-09-05). The former claim that Fourier/Vandermonde interpolation automatically proves the second line of (5.31) for $\mathcal{B}_N^{\mathrm{gen}}(I)$ is withdrawn. The exact one-mode countermodel in `v1022_telb_round3_certificates.py` has full CAR algebra M_2, whereas its even/code-generated subalgebra is diagonal; the annihilator is at distance at least 1 from that subalgebra. Vandermonde inversion reaches a full-CAR compression, but does not identify the microscopic generated algebra. Consequently the upper inclusion requires an independent FE-GEN compression-surjectivity theorem, or a precise gauge/code-projected microscopic-algebra argument. The positive phase-space/ALG1 reading is also conditional on the generated projection containing the microscopic low-energy sectors.

Theorem 5.18 (Exact reduction of L2)**.** *Assume the exhaustion estimates* (5.31)*. Then L2 of Theorem 1.1 holds.*

Proof. Lemma 5.4 gives the nonzero charged field after fixing one coherent cocycle/phase normalization. Theorem 5.12 and the integer spin $h_\lambda = 1$ make the order-four Q-system commutative. The neutral stress tensor converges by Theorem 3.1; covariance of the charged primary fields then supplies conformal covariance of the generated extension. Lemma 5.14 gives local-algebra convergence, using Proposition 5.15. Theorem 5.13 supplies the outer local action. Nonzeroness follows from the normalized vacuum-to-charge matrix element fixed to one at every scale and preserved by the graph limit. □

6 L3: finite-group C^* crossed products commute with the limit

6.1 The C^*-inductive theorem

Theorem 6.1 (Finite-group crossed-product/limit exchange)**.** *Let G be finite and let*

$$A_1 \xrightarrow{\iota_1} A_2 \xrightarrow{\iota_2} \cdots$$

be an inductive system of unital C^*-algebras with injective maps. Suppose $\alpha^{(N)} : G \curvearrowright A_N$ are actions and

$$\iota_N \alpha_g^{(N)} = \alpha_g^{(N+1)} \iota_N. \tag{6.1}$$

Let $A = \varinjlim A_N$ and $\alpha = \varinjlim \alpha^{(N)}$. Then the canonical map is a $*$-isomorphism

$$\varinjlim (A_N \rtimes_{\alpha^{(N)}} G) \cong A \rtimes_\alpha G. \tag{6.2}$$

For the coefficient expectations

$$F_N\left(\sum_{g\in G} a_g u_g\right) = a_e \tag{6.3}$$

and fixed-point expectations

$$E_N(a) = \frac{1}{|G|}\sum_{g\in G} \alpha_g^{(N)}(a), \tag{6.4}$$

the diagrams commute exactly, and their limits are the canonical expectations F and E. They are faithful. Moreover,

$$E_N(x) \geq \frac{1}{|G|}x \quad (x\geq 0), \qquad E(x) \geq \frac{1}{|G|}x \quad (x\geq 0). \tag{6.5}$$

Thus the uniform Pimsner–Popa index bound survives the limit. If each action is saturated/outer on factors, the index is exactly $|G|$.

Proof. Equivariance gives a homomorphism

$$j_N : A_N \rtimes G \longrightarrow A_{N+1} \rtimes G, \qquad j_N(a) = \iota_N(a), \quad j_N(u_g) = u_g. \tag{6.6}$$

It is injective. To see this without appealing to exactness, choose a faithful representation π_{N+1} of A_{N+1} and use the regular covariant representation on $\ell^2(G)\otimes\mathcal{H}$. Its restriction to $A_N \rtimes G$ is faithful because finite groups are amenable, so full and reduced crossed products coincide. Hence the inductive limit on the left of (6.2) is defined by injective maps.

Let $A_{\mathrm{alg}} = \bigcup_N \iota_{N,\infty}(A_N)$, dense in A. Every algebraic crossed-product element $x = \sum_{g\in G} a_g u_g$ with $a_g \in A_{\mathrm{alg}}$ lies at one common finite stage because G is finite. Therefore the union of the images of $A_N \rtimes_{\alpha^{(N)}} G$ is the algebraic crossed product $A_{\mathrm{alg}} \rtimes G$, which is dense in $A \rtimes_\alpha G$. The regular representations used above show that the norm of such an x is the same whether computed at its finite stage, in the inductive limit, or in $A \rtimes_\alpha G$. The dense isometric homomorphism therefore extends to (6.2).

On algebraic elements,

$$F_{N+1} j_N\left(\sum_g a_g u_g\right) = \iota_N(a_e) = \iota_N F_N\left(\sum_g a_g u_g\right),$$

and (6.1) gives $E_{N+1}\iota_N = \iota_N E_N$. Hence both expectations descend to the limit and agree there with (6.3) and (6.4). Faithfulness of F follows from the faithful regular representation. Faithfulness of E follows because a sum of positive operators can vanish only when every summand vanishes.

Finally, for $x \geq 0$,

$$E_N(x) = \frac{1}{|G|}\left(x + \sum_{g\neq e} \alpha_g^{(N)}(x)\right) \geq \frac{1}{|G|}x.$$

The same direct formula proves the limit inequality. For outer finite group actions on factors, the fixed-point index theorem gives equality of the index with $|G|$. $\square$

6.2 State preservation and the charged grades

Corollary 6.2 (State-preserving expectations commute with the limit)**.** *Let φ_N be compatible $\alpha^{(N)}$-invariant states. Then $\varphi_N \circ E_N = \varphi_N$, the limit state φ satisfies $\varphi \circ E = \varphi$, and the GNS extensions of E_N converge on the dense inductive union to the GNS extension of E.*

Proof. Invariance gives

$$\varphi_N(E_N(a)) = |G|^{-1}\sum_g \varphi_N(\alpha_g^{(N)}(a)) = \varphi_N(a).$$

Compatibility permits passage to the dense union. Contractivity of conditional expectations extends the equality and convergence by continuity. ☐

Corollary 6.3 ($\mathbb{Z}_4$ fusion and expectation)**.** *For $G = \mathbb{Z}_4$, the limiting homogeneous subspaces $\mathcal{B}_k = \overline{Au^k}$ satisfy*

$$\mathcal{B}_k\mathcal{B}_\ell \subset \mathcal{B}_{k+\ell}, \qquad \mathcal{B}_k^* = \mathcal{B}_{-k}, \qquad u^4 = \mathbf{1}. \tag{6.7}$$

Thus the grade-one field obeys $\Psi_\lambda^4 \in \mathcal{A}$ and $\Psi_\lambda^2 \in \mathcal{B}_2$, identified by (2.7) with the vector/parity class. The limit expectation is

$$E(x) = \frac{1}{4}\sum_{r=0}^{3}\alpha_r(x), \qquad E(x) \geq \frac{1}{4}x \quad (x \geq 0). \tag{6.8}$$

Proof. For $a, b \in A$, $(au^k)(bu^\ell) = a\alpha_k(b)u^{k+\ell}$ and $(au^k)^* = \alpha_{-k}(a^*)u^{-k}$. The remaining statements follow from Theorem 6.1. ☐

Remark 6.4 (Inner finite shadows versus outer type-III actions). At a finite matrix stage the clock action can be inner, in which case $A_N \rtimes \mathbb{Z}_4 \cong A_N \otimes C^*(\mathbb{Z}_4)$ and the inclusion need not model an irreducible outer fixed-point subfactor. The limit theorem above remains valid, but index *equality* and Q-system irreducibility require outerness/saturation of the limiting local action. Those properties belong to L2/(ALG-EXH), not to the formal crossed-product exchange.

7 L4: holomorphy and identification with $(E_8)_1$

7.1 The neutral net and its index

The intended neutral current net is

$$\mathcal{A} = \mathcal{A}_{D_5,1} \otimes \mathcal{A}_{A_3,1}. \tag{7.1}$$

Its central charge is

$$c(D_5, 1) + c(A_3, 1) = 5 + 3 = 8. \tag{7.2}$$

The discriminant groups of D_5 and A_3 each have order four, so the product lattice net has

$$\mu_{\mathcal{A}} = |\det D_5|\,|\det A_3| = 4 \cdot 4 = 16. \tag{7.3}$$

The isotropic diagonal subgroup generated by λ has order four. The exact lattice identities (2.6)–(2.8) verify integer spin, evenness and unimodularity of the extension lattice.

Theorem 7.1 (Identification once L1/L2 hold)**.** *Assume L1 and L2 produce a local irreducible index-four extension $\mathcal{A} \subset \mathcal{B}$ with the common stress tensor, with the four grades generated by λ, and with finite-dimensional L_0 eigenspaces. Then $\mathcal{B}$ is the $(E_8)_1$ conformal net.*

Proof. Complete rationality and holomorphy. The affine lattice net (7.1) is completely rational, split and strongly additive. The order-four local simple-current Q-system has dimension/index four by Theorem 3.4. KLM therefore gives

$$\mu_{\mathcal{B}} = \frac{\mu_{\mathcal{A}}}{[\mathcal{B} : \mathcal{A}]^2} = \frac{16}{4^2} = 1. \tag{7.4}$$

Hence $\mathcal{B}$ is holomorphic.

Conformal covariance and split property. The common stress tensor is the Osborne–Stottmeister limit, and L2 assumes its covariance acts on the charged primary fields. Thus the extension is conformal, not merely Möbius covariant. L1 supplies the energy-bounded generating fields. Their finite-energy spaces reconstruct a simple strongly local unitary VOA $V_{\mathcal{B}}$ by the CKLW/Fredenhagen–Jörß correspondence. The character below is finite for every $\operatorname{Im}\tau > 0$, so (3.5) holds and Theorem 3.5 supplies the split property independently.

The lattice and character. The extension lattice

$$L = \bigcup_{k=0}^{3} (D_5 \oplus A_3 + k\lambda) \tag{7.5}$$

is even by $\lambda \cdot (D_5 \oplus A_3) \subset \mathbb{Z}$ and $\|\lambda\|^2 = 2$. Its determinant is

$$\det L = \frac{\det(D_5)\det(A_3)}{4^2} = \frac{16}{16} = 1. \tag{7.6}$$

There is a unique positive-definite even unimodular lattice of rank eight, namely E_8. Therefore $L \cong E_8$. Its theta series is the unique normalized weight-four modular form,

$$\Theta_L = E_4, \tag{7.7}$$

and

$$\mathrm{ch}_{V_L} = \frac{\Theta_L}{\eta^8} = \frac{E_4}{\eta^8}. \tag{7.8}$$

The exact initial coefficients are $1, 248, 4124, 34752$; the 248 weight-one states are the 240 roots in (2.8) plus eight Cartan currents.

Uniqueness. The reconstructed VOA is simple, unitary, rational, C_2-cofinite and holomorphic: rationality and C_2-cofiniteness follow here from the finite simple-current extension of the positive-definite lattice VOA. It has $c = 8$. Theorem 3.6 gives $V_{\mathcal{B}} \cong V_{E_8}$, and CKLW identifies the generated net with $\mathcal{A}_{V_{E_8}}$, the level-one E_8 current net. □

Remark 7.2 (Which hypotheses are presently established)*.* At finite level, the order-four class, integer spin, census, determinant, character coefficients, $k = 1$ current normalization, and measured $c = 8.004666$ are established. At continuum level, the neutral fermion-bilinear currents and stress tensor lie in the Osborne–Stottmeister theorem class. What is not established is that the non-bilinear lattice charged fields satisfy L1/L2 and exhaust the local scaling algebra. Without that step, the finite E_4 character match is not a proof of conformal covariance or splitness of the collar limit.

Corollary 7.3 (Local factor type)**.** *Under Theorem 7.1, every nontrivial interval algebra $\mathcal{B}(I)$ is the hyperfinite type-*III_1 *factor.*

Proof. The E_8 level-one current net is split, irreducible, strongly additive and conformal. The standard type theorem for nontrivial conformal nets gives type III_1; splitness and the current-net construction give hyperfiniteness. □

8 Downstream corollary cascade

Corollary 8.1 (Charged-field and Ψ_λ convergence)**.** *If the final gap box is discharged, the field $\Psi_{\lambda,N}(f)$ converges on the finite-energy core, is relatively local, has order-four fusion, and generates the missing odd $64 + 64 = 128$ weight-one states. The conditional statement in Ψ_λ at* the Quasi-Free Boundary *becomes unconditional for this collar.*

Proof. Apply L1, L2 and Corollary 6.3; the state count is (2.8). □

Corollary 8.2 (Type-III KMS uniqueness). *If the final gap box is discharged, the hypotheses of the abstract holomorphic rotational KMS theorem in* The Holomorphic Rotational KMS Extension Theorem *hold for the concrete collar limit. Thus the rotational β-KMS state of the extension is the unique sector Gibbs state, and the canonical $\mathbb{Z}_4$-invariant extension of the subnet state is $\omega_\beta \circ E$.*

Proof. Theorem 7.1 gives a holomorphic conformal net with $\mu = 1$ and Corollary 7.3 gives the type-III local factors. The expectation is Corollary 6.3. KLM leaves only the vacuum DHR sector; the rotational KMS classification then leaves only its Gibbs state. □

Corollary 8.3 (P1 and Quillen premises: exact conditional scope). *If, in addition, the determinant-line modular response is identified with an additive transitive four-channel response and the angular modular normalization is $\beta_{\mathrm{angle}} = 2\pi$, then*

$$c_3 = \frac{1}{4\beta_{\mathrm{angle}}} = \frac{1}{8\pi}. \tag{8.1}$$

The KMS state has no independent sector-mixing direction, supplying the rigidity premise used in the Quillen argument.

Proof. The unique invariant state assigns equal weights to four transitively permuted channels, hence weight $1/4$ to each. Multiplying the total response $1/\beta_{\mathrm{angle}}$ by $1/4$ proves (8.1). KMS uniqueness removes a second state direction. □

Remark 8.4. The master theorem would not by itself identify the determinant-line response with the channel response. That is a separate P1/Quillen operator-identification premise. Accordingly the master theorem closes the *net/KMS rigidity premise*, not the full Quillen variation formula.

9 Per-lemma status

Item	Statement	Status	Reason
S0	Frozen generator arithmetic: $\|\lambda\|^2 = 2$, $h_\lambda = 1$, order four, census $[52, 64, 60, 64]$	PROVED	Exact rational lattice calculation; rechecked 10/10.
S1	Neutral bilinear currents and stress tensor have a scaling limit	REDUCED-TO-CITED	Osborne–Stottmeister Theorems 4.11, 4.16 and 5.1; the frozen neutral objects are in the bilinear class.
S2a	Continuum e^λ has a polynomial energy bound	PROVED	It is a norm-two, weight-one affine current; Lemma 4.1 and CKLW.
UGF	Edge-projected Green domination and local Sobolev convergence	PROVED	Theorem 4.7: exact particle–hole Schur count plus the sharp sine comparison gives $K_G = \pi^2/4$ for every N.
L1	The finite fields have an N-uniform energy bound	PROVED	Theorem 4.8, using the proved UGF and norm-one cocycle/charge shifts.
TEL-a	Dyadic edge-mode isometries and charged graph defect	PROVED	Theorem 5.3: explicit second-quantized Shannon embeddings, $p = 3$, $\epsilon = 1$, and rate N^{-2} for fields and adjoints.

Item	Statement	Status	Reason
TEL-b	Full-CAR two-edge remainder has an all-N bound	PROVED	v1022/v1025 inputs plus v1026 CF/DG, one-sided CROSS, validated Fourier core and full aliases prove $\|R_N\|_{\mathrm{HS}} < 2.995906 < 3$ for all even $N \geq 16$. Norm only; no one-edge algebra identification or historical 2.11 theorem is claimed.
L2a	Edge-projected strong convergence on the finite-energy core	PROVED	The dyadic telescoping series converges by Theorem 5.3.
ALG-rel	Relative locality with the neutral current net	PROVED	Theorem 5.12: integral root pairings plus strong and graph convergence pass the exact finite commutator to the limit.
ALG-out	Outerness of the limiting $\mathbb{Z}_4$ action	PROVED	Theorem 5.13: the connected multiplicity-free Q-system gives $\mathcal{A}(I)' \cap \mathcal{B}^{\mathrm{gen}}(I) = \mathbb{C}$.
ALG-exh	Local algebra convergence and no extra UV operators	GAP	Reduced exactly to the two Buchholz–Verch estimates (5.31).
L3a	Crossed product commutes with the inductive limit	PROVED	Theorem 6.1, proved isometrically for every finite group.
L3b	Expectations and uniform index bounds commute with the limit	PROVED	Equations (6.3)–(6.5) and Corollary 6.2.
L4a	Index-four extension has $\mu = 1$	REDUCED-TO-CITED	KLM, after L2 supplies a local irreducible conformal Q-system: $16/4^2 = 1$.
L4b	Holomorphic $c = 8$ extension is $(E_8)_1$	REDUCED-TO-CITED	Even-unimodular rank-eight classification, Dong–Mason and CKLW, after L1/L2 supply the strongly local VOA/net hypotheses.
L4c	Limit local algebras are hyperfinite type III_1	REDUCED-TO-CITED	Standard conformal-current-net factor theorem after L4b.
KMS	Unique rotational KMS extension	REDUCED-TO-CITED	The downstream theorem applies once L4 is available.
P1/Q	Equal four-channel weight and Quillen rigidity premise	REDUCED-TO-CITED	Net/KMS part follows; determinant-line response identification remains separate.

10 The remaining analytic gap

Remaining microscopic-exhaustion gaps (status 2026-09-05).
The Green-form and edge-mode graph parts are no longer assumptions: Theorems 4.7 and 5.3 prove them with $K_G = \pi^2/4$ and dyadic rate N^{-2}. Round-3 certificates close the interval-TV input, C1 and C2a. The analytic scalar part of C2b is $c_{\text{potential}} = 0.399224424139106\ldots < 0.400$. This does not yet identify the edge-mode system with the sharply localized scaling algebra of the sixteen-Majorana $M = 1$, $N_y = 8$ microscopic collar.

(TEL-B) Relaxed norm obligation closed in round 4. For every even $N \geq 16$, the now-proved estimates are

$$\sqrt{N}\ \|(K'_N - K_N)_{\text{offdiag}}\|_{\text{HS}} < 0.065697, \qquad \sqrt{N}\ \|(\text{Res}_N - K_N)_{\text{diag}}\|_{\text{HS}} < 0.042959.$$

The two components are Hilbert–Schmidt orthogonal. Combined with the certified scalar contribution, these give $c_E < 0.466902 < 0.542702$. The one-sided CROSS bound is $0.282637/N$, giving Bound-B squared < 0.714386. A 512-node Acb certificate with infinite alias control proves the weighted Fourier core < 1.783260551; the actual finite-grid smooth remainder additionally costs < 0.017333 in aliases and is < 1.800594, not the former unsupported 1.7833. With the exact jump norm $\sqrt{2}$, the unrounded assembly gives $\|R_N\|_{\text{HS}} < 2.995906 < 3$. The complete all-$N$ proof is the round-4 supplement in `tfpt_research_contracts.tex`, executed by `v1026_telb_round4_closure.py`. Neither the historical sharper 2.11 target nor the absolute-CROSS $0.283/N$ mutant is proved. The dyadic-increment route is retired, not an alternative premise.

(ALG-EXH) Buchholz–Verch upper inclusion. Prove the two estimates (5.31); equivalently, construct a support-preserving microscopic realization of the dyadic embeddings and prove, for every interval I,

$$\limsup_N \mathcal{B}_N(I) \subset W^*\Big(\mathcal{A}(I), \overline{\Psi_{k\lambda}(f)} : k \in \mathbb{Z}_4,\ \operatorname{supp} f \subset I\Big) \subset \liminf_N \mathcal{B}_N(I).$$

Equivalently, no bounded local ultraviolet sequence survives outside the neutral currents and the four charged grades. Relative locality and outerness are already Theorems 5.12 and 5.13; they are not part of this residue. The binding ALG2 correction above means that full-CAR Vandermonde interpolation does not prove this inclusion for $\mathcal{B}_N^{\text{gen}}$; FE-GEN or an exact gauge/code-projected microscopic-algebra theorem is independently required.

For the edge-projected theorem, (ALG-EXH) remains the sole hypothesis. For the original microscopic-collar formulation, the CF/DG norm inputs are now proved by the Round-4 certificate (v1026), for the fixed-width $M = 1$, $N_y = 8$ model and every even $N \geq 16$. This does not supply the microscopic-algebra identification required by FE-GEN or prove (ALG-EXH). Once (5.31) holds, L1/L2 are complete and L3 plus the cited L4 chain finish the theorem.

Conclusion

The order-four arithmetic and the finite-group operator algebra are not the obstruction. The edge-projected Green form is now uniformly dominated for every N with $K_G = \pi^2/4$, and it converges to the chiral Green form in local H^2 quadratic-form norm. Exact compatible dyadic edge-mode isometries now give the charged graph-defect rate N^{-2} for fields and adjoints. The full crossed-product/expectation limit is proved, and the holomorphic $c = 8$ endpoint is rigid once a local conformal limit with the charged field exists. The unresolved point is the support-preserving identification of this edge-mode limit with the full microscopic CAR local scaling algebra. For the original microscopic collar, v1026 now proves the relaxed uniform TEL-b norm below $2.995906 < 3$, including CF, DG and finite-grid aliases, without a finite-plateau inference. Relative locality and outerness are proved. The remaining in-house algebraic obligation is the Buchholz–Verch upper inclusion with the binding FE-GEN/gauge-code qualification. The conditional sector/actual-word frame theorem in v1030 is a concrete sufficient interface, not a proof that the microscopic one-edge algebra satisfies it. Joint-adjoint control and high-output tails remain indispensable.

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